

Mathematics Extension 2 Term 4 Assessment 2005

Question 1 (15 Marks)

Marks

- (a) Given that $\alpha = 3 + 2i$ and $\mu = 2 - 5i$, find
- (i) $\alpha\mu$ 2
 - (ii) $\operatorname{Re}(\alpha\mu)$ 1
 - (iii) $\operatorname{Im}(\alpha\mu)$ 1
- (b) Describe geometrically, on an Argand diagram
- (i) $\operatorname{Re}(z) = 2$. 1
 - (ii) $\operatorname{Im}(z) < 2$ 1
- (c) Expand and simplify $(2 - 3i)^4$ 3
- (d) Calculate the modulus of the product of the roots of the equation $(2 + i)x^2 + 3x - (1 - i) = 0$. 3
- (e) Show that the point representing $\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ is on a circle with radius 1 and centre at (1, 0). 3

Question 2 (15 Marks)

- (a) The complex number $z = \sqrt{3} + i$ is represented on an Argand diagram by the point A.
- (i) The points B, C, D and E are the points representing $-z, iz, 1 - z$ and $\bar{z}$ respectively. Mark clearly on an Argand diagram the points A, B, C, D , and E . Clearly indicate important geometrical relationships between these points. 4
 - (ii) F is the point in the second quadrant such that $\triangle ABF$ is equilateral. What complex number is represented by the point F ? 2
- (b) (i) Find $\sqrt{6i - 8}$, in the form $a + ib$. 3
- (ii) Hence, solve the equation $2z^2 - (3 + i)z + 2 = 0$, expressing the values of z in the form $a + ib$. 2
- (c) A point z on the Argand diagram is given by $z = w^2 + 2iw$, where $w = u + iv$ and u and v are real. Find the locus of z when
- (i) $u = 0$ and v varies. 1
 - (ii) $v = 1$ and u varies. 1
 - (iii) Sketch the two loci, showing any important features. 2

Mathematics Extension 2 Term 4 Assessment 2005

Question 3 (15 Marks)	Marks
(a) Sketch the locus $ z + 2 - 3i \leq 5$.	2
(b) Show that $w = 2\sqrt{3}i - 2$ is a root of the equation $z^3 = 64$.	2
(c) If $z \neq 0$, show that $u = z + \frac{ z ^2}{z}$ is always real, where $z = x + iy$ and $x \in \mathbb{R}$, $y \in \mathbb{R}$.	3
(d) If $z = \cos\theta + i\sin\theta$, prove that $\frac{2}{1+z} = 1 - i \tan \frac{\theta}{2}$.	3
(e) Sketch the region, in an Argand diagram where points satisfy the set of inequalities: $2 \leq z \leq 4$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$.	2
(f) Prove that if Z_1, Z_2 are complex numbers then $ Z_1 - Z_2 ^2 + Z_1 + Z_2 ^2 = 2\{ Z_1 ^2 + Z_2 ^2\}$.	3

Question 4 (15 Marks)

(a) $2 - i$ is one root of $x^2 - (3 - i)x + k = 0$. Find k and the other root of the equation.	2
(b) Sketch the region in the Argand diagram defined by $ z^2 - \bar{z}^2 \geq 8$	3
(c) (i) Show that $(1 + i)$ is a root of the polynomial $P(x) = x^3 + x^2 - 4x + 6$.	2
(ii) Hence resolve $P(x)$ into irreducible factors over the complex field.	3
(d) Find the fourth roots of $2\sqrt{3} + 2i$.	5

 END OF EXAM 

Question 1 (15 Marks)

Marks

(a) Given that $\alpha = 3 + 2i$ and $\mu = 2 - 5i$, find

(i) $\alpha\mu = (3 + 2i)(2 - 5i)$ 1mk
 $= 11 - 4i$ 1mk

2

(ii) $\operatorname{Re}(\alpha\mu) = 11$ 1mk

1

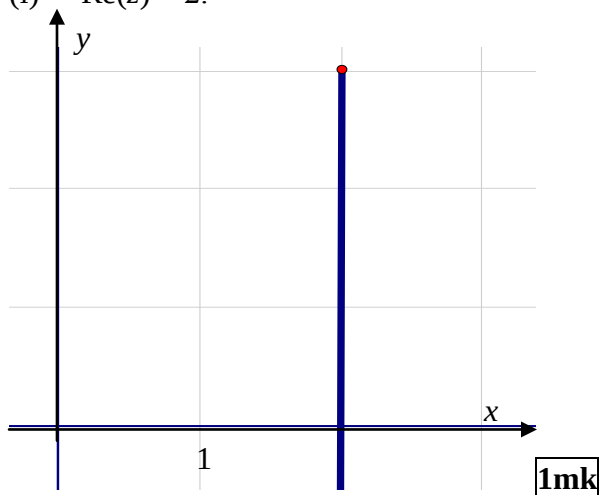
(iii) $\operatorname{Im}(\alpha\mu) = -4$ 1mk

1

(b) Describe geometrically, on an Argand diagram

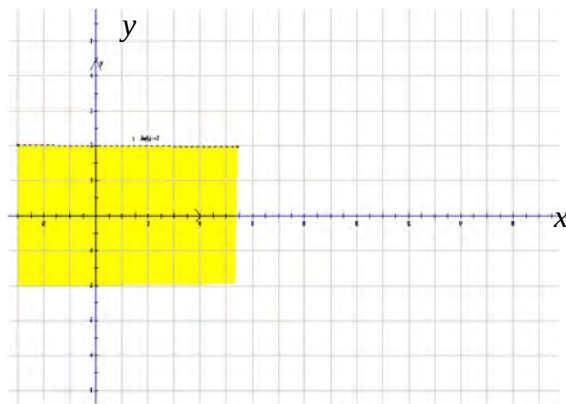
(i) $\operatorname{Re}(z) = 2$.

1



(ii) $\operatorname{Im}(z) < 2$

1



1mk

(c) Expand and simplify $(2 - 3i)^4$

3

$(2 - 3i)^4 = 2^4 + 4(2)^3(-3i) + 6(2)^2(-3i)^2 + 4(2)(-3i)^3 + (-3i)^4$ 1mk
 $= 16 - 96i + 216i^2 - 216i^3 + 81i^4$ 1mk
 $= -119 + 120i$ 1mk

- (d) Calculate the modulus of the product of the roots of the equation $(2+i)x^2 + 3x - (1-i) = 0$.

3

Let the roots be α, β . Then $\alpha\beta = \frac{-1+i}{2+i}$ 1mk

$$\therefore \alpha\beta = \frac{-1+i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{-3+3i}{5}$$
1mk

$$|\alpha\beta| = \sqrt{\left(\frac{-3}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \frac{3\sqrt{2}}{5}$$
1mk

- (e) Show that the point representing $\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ is on a circle with radius 1 and centre at (1, 0).

3

Let $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} = \frac{1}{2} + i \frac{\sqrt{3}}{2}$

$$\therefore |z| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 \rightarrow \text{equal to radius of circle}$$
1mk

Now equation of the circle with centre (1,0) and radius 1 is $(x-1)^2 + y^2 = 1$ 1mk

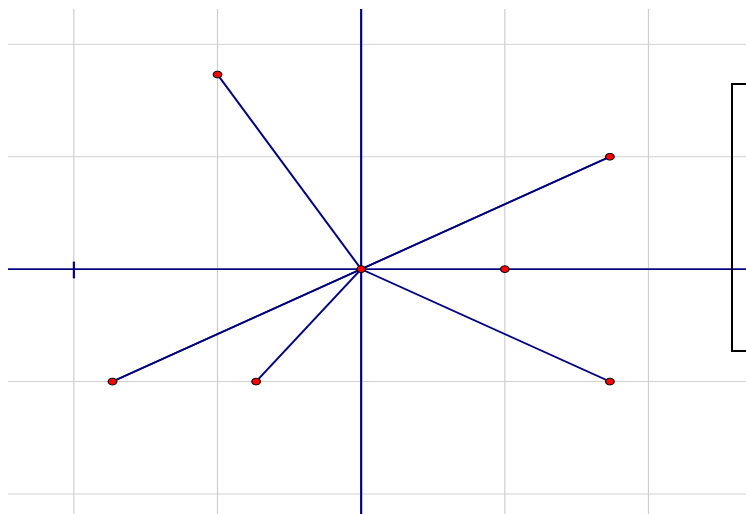
Since $x = \frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$ and substitution into LHS $= \left(\frac{-1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = 1 = \text{RHS}$ 1mk

$\therefore \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ lies on circle centred (1, 0) and radius 1.

Question 2 (15 Marks)

- (a) The complex number $z = \sqrt{3} + i$ is represented on an Argand diagram by the point A.
- (i) The points B, C, D and E are the points representing $-z$, iz , $1-z$ and $\bar{z}$ respectively. Mark clearly on an Argand diagram the points A, B, C, D, and E. Clearly indicate important geometrical relationships between these points.

4



- Points B, C, D, E with relationship to point A, award 1mk each
- Just points award 2 marks only

Mathematics Extension 2 Term 4 Assessment 2005 - solutions

- (ii) F is the point in the second quadrant such that $\triangle ABF$ is equilateral. 2
What complex number is represented by the point F ?

F must be the same distance from A as B is so must have point $(-\sqrt{3}, 2)$

So has the complex number $z_2 = -\sqrt{3} + 2i$ 1mk

- (b) (i) Find $\sqrt{6i-8}$, in the form $a + ib$. 3

Let $\Delta = 6i - 8$ then let $\Delta = (a + ib)^2$ where a, b are Real

$\therefore a^2 - b^2 = -8$ and $2ab = 6 \therefore ab = 3$ 1mk

$\therefore a^2 - \frac{9}{a^2} = -8 \rightarrow a^4 + 8a^2 - 9 = 0$

$(a^2 + 9)(a^2 - 1) = 0$ since a is real $\rightarrow a = 1$ or $a = -1$ so $b = 3$ or $b = -3$ respectively. 1mk

$\therefore \sqrt{6i-8} = 1 + 3i$ or $-1 - 3i$ 1mk

- (ii) Hence, solve the equation $2z^2 - (3 + i)z + 2 = 0$, expressing the values of z in the form $a + ib$. 2

$\therefore z = \frac{(3+i) \pm (1+3i)}{4}$ by quadratic formula 1mk

$\therefore z = 1 + i$ or $z = \frac{1}{2} - \frac{i}{2}$ 1mk

- (c) A point z on the Argand diagram is given by $z = w^2 + 2iw$, where $w = u + iv$ and u and v are real. Find the locus of z when

- (i) $u = 0$ and v varies. 1

when $u = 0$ then $w = iv$ and $z = -v^2 - 2v$

if $z = x + iy$ where x, y are real $\therefore x = -v^2 - 2v$ and $y = 0$

$\therefore$ locus is $y = 0$ 1mk

- (ii) $v = 1$ and u varies. 1

when $v = 1$ then $w = u + i$ and $z = (u + i)^2 + 2i(u + i)$

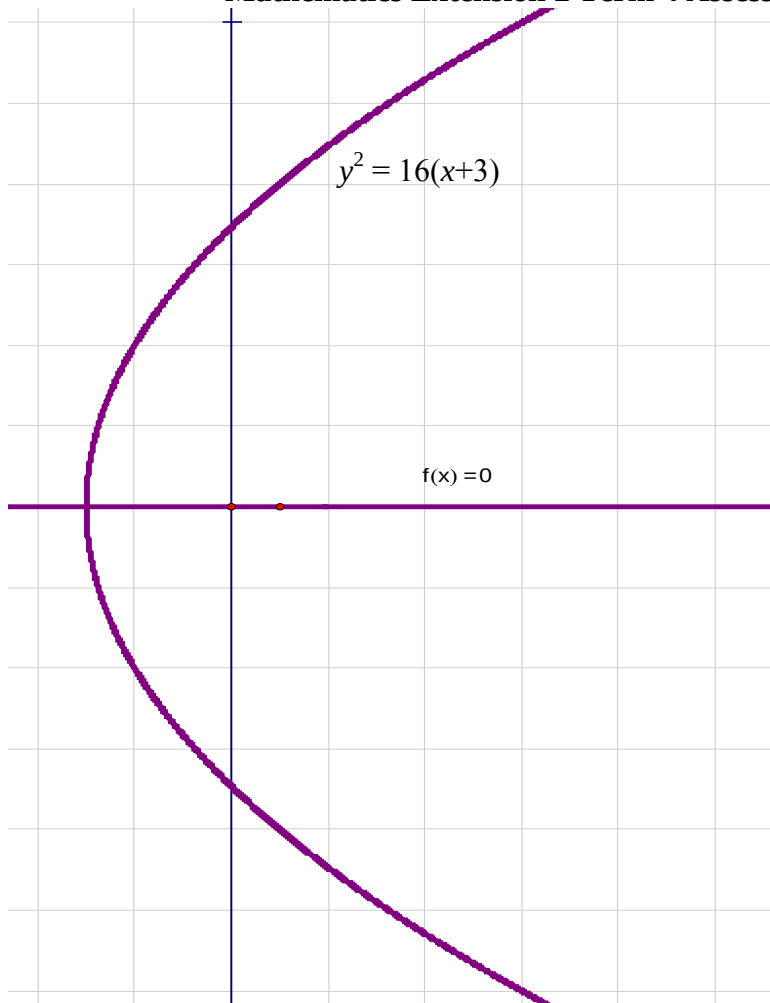
$\therefore z = (u^2 - 3) + 4ui$

$\therefore$ if $z = x + iy$ where x, y are real then

$x = u^2 - 3$ and $y = 4u \therefore y^2 = 16(x + 3)$ 1mk

- (iii) Sketch the two loci, showing any important features. 2

See next page 1mk each

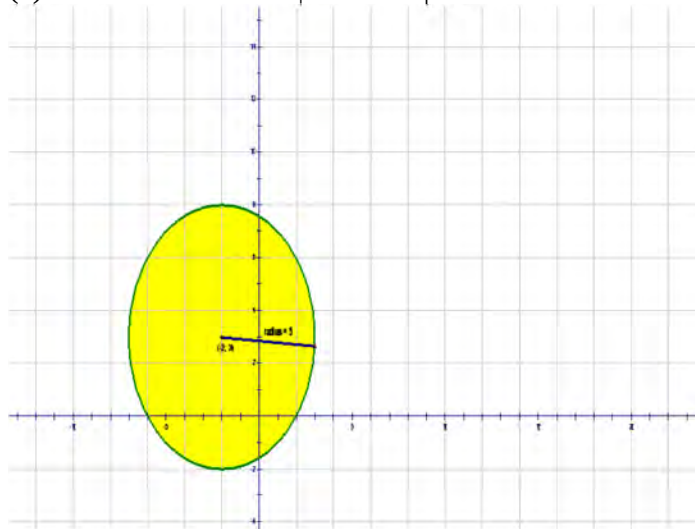


Question 3 (15 Marks)

Marks

- (a) Sketch the locus $|z + 2 - 3i| \leq 5$.

2



1mk for circle centre $(-2,3)$
and radius= 5

1mk for correct region

Mathematics Extension 2 Term 4 Assessment 2005 - solutions

- (b) Show that $w = 2\sqrt{3}i - 2$ is a root of the equation $z^3 = 64$. 2

$$w = 2\sqrt{3}i - 2 = 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) \quad \boxed{1\text{mk}}$$

$$\therefore w^3 = 4^3 (\cos 2\pi + i\sin 2\pi) \text{ by De Moivre's Theorem} \\ = 64 \text{ as required} \quad \boxed{1\text{mk}}$$

- (c) If $z \neq 0$, show that $u = z + \frac{|z|^2}{z}$ is always real, where $z = x + iy$ and $x \in \mathbb{R}$, $y \in \mathbb{R}$. 3

$$z = x + iy \text{ then } |z|^2 = x^2 + y^2 \quad \& \quad z^2 = (x^2 - y^2) + 2ixy \quad \boxed{1\text{mk}}$$

$$\text{Now } u = z + \frac{|z|^2}{z} = \frac{x^2 - y^2 + 2ixy + x^2 + y^2}{x + iy} \quad \boxed{1\text{mk}} \\ = \frac{2x(x + iy)}{x + iy} \quad \boxed{1\text{mk}} \\ = 2x \quad \text{which is real as } x \text{ is real}$$

Alternatively:

$$u = z + \frac{|z|^2}{z} = z + \frac{|z|^2 \times \bar{z}}{z \times \bar{z}} \quad \boxed{1\text{mk}} \\ = z + \frac{|z|^2 \times \bar{z}}{(\bar{z})^2} \quad \boxed{1\text{mk}} \\ = z + \bar{z} \quad \boxed{1\text{mk}} \\ = x + iy + x - iy \\ = 2x \quad \text{which is real as } x \text{ is real.}$$

- (d) If $z = \cos\theta + i\sin\theta$, prove that $\frac{2}{1+z} = 1 - i \tan \frac{\theta}{2}$. 3

$$\text{LHS} = \frac{2}{1 + \cos\theta + i\sin\theta} \times \frac{(1 + \cos\theta) - i\sin\theta}{(1 + \cos\theta) - i\sin\theta} \quad \boxed{1\text{mk}}$$

$$= \frac{2 + 2\cos\theta - 2i\sin\theta}{1 + 2\cos\theta + \cos^2\theta + \sin^2\theta}$$

$$= \frac{2 + 2\cos\theta - 2i\sin\theta}{2 + 2\cos\theta}$$

$$= 1 - \frac{i\sin\theta}{1 + \cos\theta} \quad \boxed{1\text{mk}}$$

$$= 1 - \frac{2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{1 + 2\cos^2\frac{\theta}{2} - 1} \quad \boxed{1\text{mk}}$$

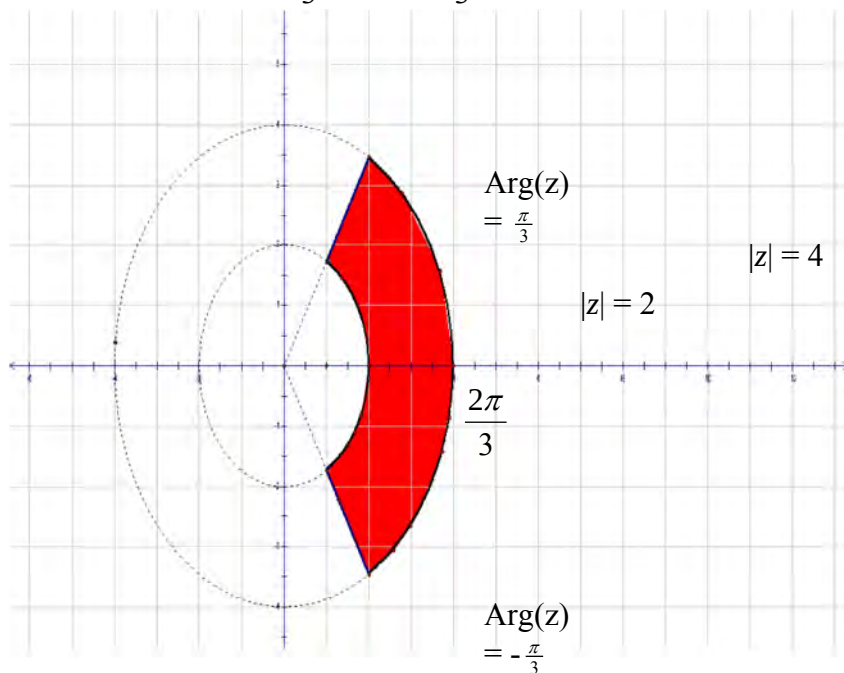
$$= 1 - \frac{i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}$$

$$= 1 - i \tan \frac{\theta}{2} = \text{RHS} \quad \left[\text{Students could also let } t = \tan \frac{\theta}{2} \right]$$

Mathematics Extension 2 Term 4 Assessment 2005 - solutions

- (e) Sketch the region, in an Argand diagram where points satisfy the set of inequalities: 2

$$2 \leq |z| \leq 4 \text{ and } -\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}.$$



1mk for the boundary

1mk for the region

- (f) Prove that if Z_1, Z_2 are complex numbers then $|Z_1 - Z_2|^2 + |Z_1 + Z_2|^2 = 2\{|Z_1|^2 + |Z_2|^2\}$. 3

$$\begin{aligned} \text{LHS} &= |Z_1 - Z_2|^2 + |Z_1 + Z_2|^2 \\ &= (Z_1 - Z_2)(\overline{Z_1 - Z_2}) + (Z_1 + Z_2)(\overline{Z_1 + Z_2}) \quad \text{1mk} \\ &= (Z_1 - Z_2)(\overline{Z_1} - \overline{Z_2}) + (Z_1 + Z_2)(\overline{Z_1} + \overline{Z_2}) \\ &= 2(Z_1\overline{Z_1} + Z_2\overline{Z_2}) \quad \text{1mk} \\ &= 2(|Z_1|^2 + |Z_2|^2) = \text{RHS} \quad \text{1mk} \end{aligned}$$

Question 4 (15 Marks)

- (a) $2 - i$ is one root of $x^2 - (3 - i)x + k = 0$. Find k and the other root of the equation. 2

Let α and β be the roots of the equation, so let $\beta = 2 - i$

$$\therefore \alpha + 2 - i = (3 - i) \rightarrow \alpha = 1 \quad \text{1mk}$$

$$\therefore \alpha\beta = k \rightarrow k = 2 - i \quad \text{1mk}$$

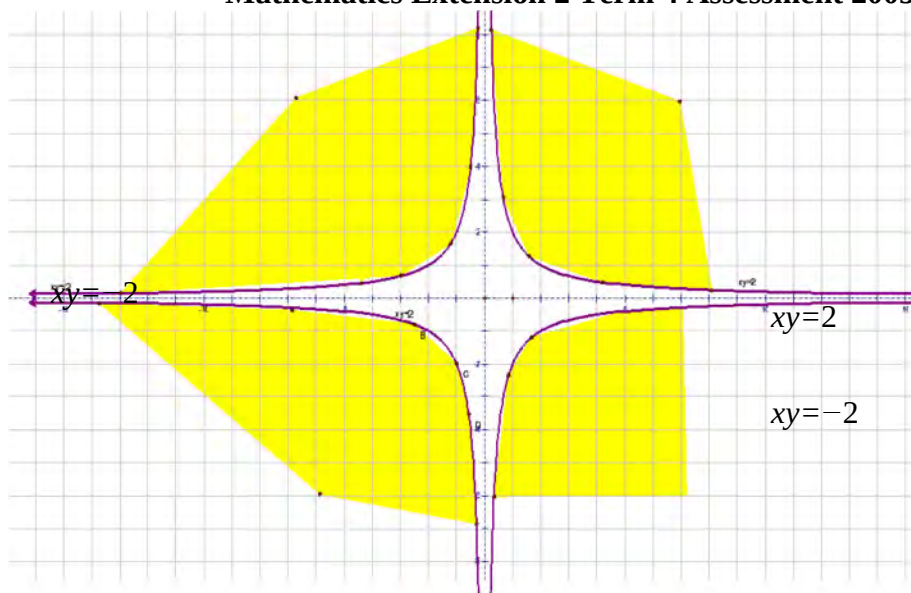
- (b) Sketch the region in the Argand diagram defined by $|z^2 - \bar{z}^2| \geq 8$ 3

$$\begin{aligned} z^2 - \bar{z}^2 &= (z - \bar{z})(z + \bar{z}) \\ &= (2iy)(2x) = 4ixy \end{aligned}$$

$$\therefore |z^2 - \bar{z}^2| \geq 4|xy|$$

$$\therefore 4|xy| \geq 8 \rightarrow |xy| \geq 2 \quad \text{1mk}$$

Mathematics Extension 2 Term 4 Assessment 2005 - solutions



1 mk for the graphs
and 1 mk for the
correct region.

- (c) (i) Show that $(1 + i)$ is a root of the polynomial $P(x) = x^3 + x^2 - 4x + 6$. 2

If $(1 + i)$ is a root then $P(1 + i) = 0$

$$\begin{aligned} P(1 + i) &= (1 + i)^3 + (1 + i)^2 - 4(1 + i) + 6 \\ &= 1 + 3i + 3i^2 + i^3 + 1 + 2i + i^2 - 4 - 4i + 6 \\ &= 4 + i - 3 - 1 - i = 0 \end{aligned} \quad \boxed{1\text{mk}} \quad \boxed{1\text{mk}}$$

$\therefore (1 + i)$ is a root of $P(x)$.

- (ii) Hence resolve $P(x)$ into irreducible factors over the complex field. 3

Since $P(x)$ has real coefficients $\rightarrow$ the complex roots occur in conjugate pairs.

$\therefore (1 - i)$ is also a root. 1mk

$\therefore$ Let $P(x)$ have roots $(1 + i)$, $(1 - i)$ and β

Sum of the roots $= (1 + i) + (1 - i) + \beta = -1 \rightarrow \beta = -3$ 1mk

$\therefore P(x) = (x + 3) \{x - (1 + i)\} \{x - (1 - i)\}$ 1mk

- (d) Find the fourth roots of $2\sqrt{3} + 2i$. 5

Let $z = x + iy$ be one of the fourth roots. Then $z^4 = 2\sqrt{3} + 2i$

$\therefore z^4 = 16(\cos 4\theta + i \sin 4\theta)$ by De Moivre's Theorem.

Equating the real parts $\rightarrow 16\cos 4\theta = 2\sqrt{3}$ 1mk

Equating imaginary parts $\rightarrow 16\sin 4\theta = 2$ 1mk

$$\therefore \tan 4\theta = \frac{1}{\sqrt{3}} \quad \boxed{1\text{mk}} \quad \rightarrow \quad 4\theta = \frac{\pi}{6} + 2k\pi, \text{ where } k \text{ are integers.}$$

$$\therefore \theta = \frac{(12k+1)\pi}{24}, \text{ where } k = 0, 1, -1, -2 \quad \boxed{1\text{mk}}$$

$$\therefore \text{Four roots } \sqrt{3} + i \text{ are: } 2\text{cis} \frac{\pi}{24}, 2\text{cis} \frac{13\pi}{24}, 2\text{cis} \frac{-11\pi}{24}, 2\text{cis} \frac{-23\pi}{24} \quad \boxed{1\text{mk}}$$

END OF EXAM ☺